Evaluation runs: paxos (paxos-oopsla25)

generated 2026-11-05 from brain-data cfc25f174 2026-11-05 13:40:01 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair 1b-none, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	15	268992	conforms	2.6	4	1 h, 250 GB	2	12	brain03, 2026-11-02
2	15	455616	violation	31.5	10	1 h, 250 GB	186626	2799371	brain03, 2026-11-02
3	15	828864	violation	99.0	22	1 h, 250 GB	559874	8398091	brain03, 2026-11-02

stateless: pair correct, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	15	268992	conforms	2.6	2	1 h, 250 GB	3	16	brain03, 2026-11-02
2	15	455616	conforms	4.5	4	1 h, 250 GB	3	22	brain03, 2026-11-02
3	15	828864	conforms	8.9	6	1 h, 250 GB	3	21	brain03, 2026-11-02

stateless: pair half-majority, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	15	416520	conforms	2.8	4	1 h, 250 GB	2	12	brain03, 2026-11-02
2	15	795600	violation	25.3	8	1 h, 250 GB	168482	2527212	brain03, 2026-11-02
3	15	1553760	violation	86.0	12	1 h, 250 GB	505442	7581611	brain03, 2026-11-02

stateless: pair ignore-max-vote, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	15	268992	conforms	2.6	2	1 h, 250 GB	2	12	brain03, 2026-11-02
2	15	455616	violation	32.3	6	1 h, 250 GB	186626	2799371	brain03, 2026-11-02
3	15	828864	violation	101.2	4	1 h, 250 GB	559874	8398095	brain03, 2026-11-02

stateless: pair skip-2a-check, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	15	1962432	conforms	11.4	6	1 h, 250 GB	3	18	brain03, 2026-11-02
2	15	3921984	violation	122.4	16	1 h, 250 GB	870915	13063701	brain03, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
3	15	7841088	violation	436.6	46	1 h, 250 GB	2612739	39191058	brain03, 2026-11-02

Not run (stateless). the tool ended with an error (exit-101): 4 half-majority bag, 4 ignore-max-vote bag, 4 correct bag, 4 1b-none bag, 4 skip-2a-check bag.

skipped: a smaller size had reached its limit: 5 ignore-max-vote bag, 5 correct bag, 5 half-majority bag, 5 1b-none bag, 5 skip-2a-check bag, 6 correct bag, 6 ignore-max-vote bag, 6 half-majority bag, 6 1b-none bag, 6 skip-2a-check bag, 7 correct bag, 7 half-majority bag, 7 ignore-max-vote bag, 7 1b-none bag, 7 skip-2a-check bag, 8 correct bag, 8 ignore-max-vote bag, 8 half-majority bag, 8 1b-none bag, 8 skip-2a-check bag.

stateful

stateful: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	15	268992	conforms	2.8	2	1 h, 250 GB	brain02, 2026-11-01
2	15	455616	violation	5.2	4	1 h, 250 GB	brain02, 2026-11-01
3	15	828864	violation	10.4	6	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	15	268992	conforms	2.9	4	1 h, 250 GB	brain02, 2026-11-01
2	15	455616	conforms	5.3	2	1 h, 250 GB	brain02, 2026-11-01
3	15	828864	conforms	10.7	6	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	15	416520	conforms	2.9	2	1 h, 250 GB	brain02, 2026-11-01
2	15	795600	violation	6.5	4	1 h, 250 GB	brain02, 2026-11-01
3	15	1553760	violation	12.7	6	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	15	268992	conforms	2.9	4	1 h, 250 GB	brain02, 2026-11-01
2	15	455616	violation	5.5	2	1 h, 250 GB	brain02, 2026-11-01
3	15	828864	violation	10.3	4	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	15	1962432	conforms	12.6	6	1 h, 250 GB	brain02, 2026-11-01
2	15	3921984	violation	26.6	10	1 h, 250 GB	brain02, 2026-11-01
3	15	7841088	violation	61.3	22	1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). the tool ended with an error (exit-101): 4 half-majority bag, 4 ignore-max-vote bag, 4 correct bag, 4 1b-none bag, 4 skip-2a-check bag.

skipped: a smaller size had reached its limit: 5 ignore-max-vote bag, 5 correct bag, 5 half-majority bag, 5 1b-none bag, 5 skip-2a-check bag, 6 correct bag, 6 ignore-max-vote bag, 6 half-majority bag, 6 1b-none bag, 6 skip-2a-check bag, 7 correct bag, 7 half-majority bag, 7 ignore-max-vote bag, 7 1b-none bag, 7 skip-2a-check bag, 8 correct bag, 8 ignore-max-vote bag, 8 half-majority bag, 8 1b-none bag, 8 skip-2a-check bag.

offline

offline: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	3.1	4	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain05, 2026-11-02
2	15	455616	violation	5.8	4	10 min, 16 GB	0.1	5.8	0.0	in-memory	brain05, 2026-11-02
3	15	828864	violation	11.5	25	10 min, 16 GB	0.0	11.5	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	3.3	4	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain05, 2026-11-02
2	15	455616	conforms	6.1	4	10 min, 16 GB	0.1	6.1	0.0	in-memory	brain05, 2026-11-02
3	15	828864	conforms	11.6	25	10 min, 16 GB	0.0	11.6	0.0	in-memory	brain05, 2026-11-02

offline: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	416520	conforms	3.6	4	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain05, 2026-11-02
2	15	795600	violation	6.7	6	10 min, 16 GB	0.1	6.7	0.0	in-memory	brain05, 2026-11-02
3	15	1553760	violation	14.2	25	10 min, 16 GB	0.0	14.2	0.0	in-memory	brain05, 2026-11-02

offline: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	3.2	4	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain05, 2026-11-02
2	15	455616	violation	5.9	4	10 min, 16 GB	0.1	5.9	0.0	in-memory	brain05, 2026-11-02
3	15	828864	violation	11.3	25	10 min, 16 GB	0.0	11.3	0.0	in-memory	brain05, 2026-11-02

offline: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	1962432	conforms	1629.2	25	10 min, 16 GB	0.1	1629.2	0.0	in-memory	brain05, 2026-11-02
2	15	3921984	violation	28.0	27	10 min, 16 GB	0.1	28.0	0.0	in-memory	brain05, 2026-11-02
3	15	7841088	violation	60.1	34	10 min, 16 GB	0.0	60.1	0.0	in-memory	brain05, 2026-11-02

Not run (offline). skipped: a smaller size had reached its limit: 4 ignore-max-vote bag, 4 half-majority bag, 4 correct bag, 4 1b-none bag, 4 skip-2a-check bag, 5 ignore-max-vote bag, 5 correct bag, 5 half-majority bag, 5 1b-none bag, 5 skip-2a-check bag, 6 correct bag, 6 ignore-max-vote bag, 6 half-majority bag, 6 1b-none bag, 6 skip-2a-check bag, 7 correct bag, 7 half-majority bag, 7 ignore-max-vote bag, 7 1b-none bag, 7 skip-2a-check bag, 8 correct bag, 8 ignore-max-vote bag, 8 half-majority bag, 8 1b-none bag, … (25 instances in all).

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	309.6	20844	10 min, 16 GB	0.1	309.6	0.0	in-memory	brain01, 2026-11-02
2	15	455616	violation	592.4	23134	10 min, 16 GB	0.1	592.4	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	315.3	19592	10 min, 16 GB	0.1	315.3	0.0	in-memory	brain01, 2026-11-02
2	15	455616	conforms	601.7	22698	10 min, 16 GB	0.1	601.7	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	416520	conforms	398.9	26396	10 min, 16 GB	0.1	398.9	0.0	in-memory	brain01, 2026-11-02
2	15	795600	violation	816.6	35184	10 min, 16 GB	0.1	816.6	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	356.5	22754	10 min, 16 GB	0.1	356.5	0.0	in-memory	brain01, 2026-11-02
2	15	455616	violation	660.3	26315	10 min, 16 GB	0.1	660.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	1962432	conforms	2189.0	45227	10 min, 16 GB	0.1	2189.0	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). skipped: a smaller size had reached its limit: 2 bag skip-2a-check.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	3.1	170	10 min, 16 GB	2.9	3.1	0.0	in-memory	brain01, 2026-11-02
2	15	455616	violation	5.8	168	10 min, 16 GB	3.1	5.8	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	3.3	170	10 min, 16 GB	2.9	3.3	0.0	in-memory	brain01, 2026-11-02
2	15	455616	conforms	6.1	168	10 min, 16 GB	3.1	6.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	416520	conforms	3.6	170	10 min, 16 GB	2.9	3.6	0.0	in-memory	brain01, 2026-11-02
2	15	795600	violation	6.7	168	10 min, 16 GB	3.1	6.7	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	268992	conforms	3.2	170	10 min, 16 GB	2.9	3.2	0.0	in-memory	brain01, 2026-11-02
2	15	455616	violation	5.9	168	10 min, 16 GB	3.1	5.9	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	15	1962432	conforms	1629.2	170	10 min, 16 GB	2.9	1629.2	0.0	in-memory	brain01, 2026-11-02
2	15	3921984	violation	28.0	168	10 min, 16 GB	3.1	28.0	0.0	in-memory	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	15	268992	conforms	3.3	4	1 h, 250 GB		brain01, 2026-11-02
2	15	455616	violation	5.9	4	1 h, 250 GB	305	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	15	268992	conforms	3.4	4	1 h, 250 GB		brain01, 2026-11-02
2	15	455616	conforms	6.6	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	15	416520	conforms	3.6	4	1 h, 250 GB		brain01, 2026-11-02
2	15	795600	violation	6.8	6	1 h, 250 GB	57	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	15	268992	conforms	3.3	4	1 h, 250 GB		brain01, 2026-11-02
2	15	455616	violation	6.0	4	1 h, 250 GB	333	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	15	1962432	conforms	16.1	24	1 h, 250 GB		brain01, 2026-11-02
2	15	3921984	violation	32.6	29	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992	conforms	309.6	20844	10 min, 16 GB	309.6	3.1	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	conforms	592.4	23134	10 min, 16 GB	592.4	5.8	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992	conforms	315.3	19592	10 min, 16 GB	315.3	3.3	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	conforms	601.7	22698	10 min, 16 GB	601.7	6.1	0.0	in-memory	brain01, 2026-11-02
3	828864	828864	conforms	1211.1	24622	48 h, 250 GB	1211.1	12.7	0.0	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	416520	416520	conforms	398.9	26396	10 min, 16 GB	398.9	3.6	0.0	in-memory	brain01, 2026-11-02
2	795600	795600	conforms	816.7	35184	10 min, 16 GB	816.6	6.7	0.1	in-memory	brain01, 2026-11-02
3	1553760	1553760	conforms	1687.1	39622	48 h, 250 GB	1687.1	16.2	0.0	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992	conforms	356.5	22754	10 min, 16 GB	356.5	3.2	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	conforms	660.4	26315	10 min, 16 GB	660.3	5.9	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1962432	1962432	conforms	2189.1	45227	10 min, 16 GB	2189.0	1629.2	0.0	in-memory	brain01, 2026-11-02
3	7841088	7841088	conforms	10195.0	81218	48 h, 250 GB	10195.0	75.0	0.0	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **1b-none**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992		conforms*	315.3	19592	10 min, 16 GB	315.3	3.1	0.0	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	455616	455616	2	violation	601.7	22698	10 min, 16 GB	601.7	5.8	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **half-majority**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	416520		conforms*	315.3	19592	10 min, 16 GB	315.3	3.6	0.0	in-memory	brain01, 2026-11-02
2	455616	795600	4	violation	601.7	22698	10 min, 16 GB	601.7	6.7	0.0	in-memory	brain01, 2026-11-02
3	828864	1553760	12	violation	1211.1	24622	48 h, 250 GB	1211.1	16.2	0.0	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **ignore-max-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992		conforms*	315.3	19592	10 min, 16 GB	315.3	3.2	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	2	violation	601.7	22698	10 min, 16 GB	601.7	5.9	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **skip-2a-check**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	1962432		conforms*	1629.2	19592	10 min, 16 GB	315.3	1629.2	0.0	in-memory	brain01, 2026-11-02
2	455616	3921984	2	violation	601.7	22698	10 min, 16 GB	601.7	28.0	0.0	in-memory	brain01, 2026-11-02
3	828864	7841088	6	violation	1211.1	24622	48 h, 250 GB	1211.1	75.0	0.0	merge	brain01, 2026-11-04

Not run (twins offline, Rust implementation against its Java twin). refused (an input missing at run time): 2 bag skip-2a-check.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992	conforms	309.6	20844	10 min, 16 GB	3.1	309.6	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	conforms	592.4	23134	10 min, 16 GB	5.8	592.4	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992	conforms	315.3	19592	10 min, 16 GB	3.3	315.3	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	conforms	601.7	22698	10 min, 16 GB	6.1	601.7	0.0	in-memory	brain01, 2026-11-02
3	828864	828864	conforms	1211.1	24622	48 h, 250 GB	12.7	1211.1	0.0	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	416520	416520	conforms	398.9	26396	10 min, 16 GB	3.6	398.9	0.0	in-memory	brain01, 2026-11-02
2	795600	795600	conforms	816.6	35184	10 min, 16 GB	6.7	816.6	0.0	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	1553760	1553760	conforms	1687.1	39622	48 h, 250 GB	16.2	1687.1	0.0	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **ignore-max-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992	conforms	356.5	22754	10 min, 16 GB	3.2	356.5	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	conforms	660.3	26315	10 min, 16 GB	5.9	660.3	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1962432	1962432	conforms	2189.0	45227	10 min, 16 GB	1629.2	2189.0	0.0	in-memory	brain01, 2026-11-02
3	7841088	7841088	conforms	10195.0	81218	48 h, 250 GB	75.0	10195.0	0.0	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **1b-none**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992		conforms*	309.6	20844	10 min, 16 GB	3.3	309.6	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	2	violation	592.4	23134	10 min, 16 GB	6.1	592.4	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **half-majority**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	416520		conforms*	398.9	26396	10 min, 16 GB	3.3	398.9	0.0	in-memory	brain01, 2026-11-02
2	455616	795600	4	violation	816.6	35184	10 min, 16 GB	6.1	816.6	0.0	in-memory	brain01, 2026-11-02
3	828864	1553760	12	violation	1687.1	39622	48 h, 250 GB	12.7	1687.1	0.0	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **ignore-max-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	268992		conforms*	356.5	22754	10 min, 16 GB	3.3	356.5	0.0	in-memory	brain01, 2026-11-02
2	455616	455616	2	violation	660.3	26315	10 min, 16 GB	6.1	660.3	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **skip-2a-check**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	268992	1962432		conforms*	2189.0	45227	10 min, 16 GB	3.3	2189.0	0.0	in-memory	brain01, 2026-11-02
3	828864	7841088	6	violation	10195.0	81218	48 h, 250 GB	12.7	10195.0	0.0	merge	brain01, 2026-11-04

Not run (twins offline, Java implementation against its Rust twin). skipped: a smaller size had reached its limit: 2 bag skip-2a-check.

twins online, Rust implementation (TraceForge) against the Java twin's export

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	828864	828864	conforms	13.3	25	48 h, 250 GB		brain01, 2026-11-04

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	1553760	1553760	conforms	18.0	25	48 h, 250 GB		brain01, 2026-11-04

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **skip-2a-check**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	7841088	7841088	conforms	86.5	33	48 h, 250 GB		brain01, 2026-11-04

Not run (twins online, Rust implementation (TraceForge) against the Java twin's export). planned but never scheduled (grid points beyond the ladder): 1 bag ignore-max-vote, 1 bag correct, 1 bag half-majority, 1 bag 1b-none, 1 bag skip-2a-check, 2 bag correct, 2 bag ignore-max-vote, 2 bag half-majority, 2 bag 1b-none, 2 bag skip-2a-check.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **1b-none**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	268992					1 h, 250 GB		brain02, 2026-11-02
2	455616					1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	268992					1 h, 250 GB		brain02, 2026-11-02
2	455616					1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **half-majority**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	416520					1 h, 250 GB		brain02, 2026-11-02
2	795600					1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair ignore-max-vote, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	268992					1 h, 250 GB		brain02, 2026-11-02
2	455616					1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair skip-2a-check, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1962432					1 h, 250 GB		brain02, 2026-11-02
2	3921984					1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair 1b-none, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	268992						1 h, 250 GB		brain02, 2026-11-02
2	455616						1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair half-majority, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	268992						1 h, 250 GB		brain02, 2026-11-02
2	455616						1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair ignore-max-vote, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	268992						1 h, 250 GB		brain02, 2026-11-02
2	455616						1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair skip-2a-check, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	268992						1 h, 250 GB		brain02, 2026-11-02
2	455616						1 h, 250 GB		brain02, 2026-11-02